Data and regression

Last time: class activity

$$\text{performance} = \beta_0 + \beta_1\text{size} + \beta_2\text{females} + \beta_3\text{diversity} + \beta_4\text{testosterone} \\ + \beta_5\text{diversity} \cdot \text{testosterone} + \varepsilon$$

The researchers hypothesize that increased group diversity is positively associated with performance when group testosterone levels are low, but increased diversity is negatively associated with performance when testosterone levels are high.

> What parameter are we interested in for statistical inference?

Last time: class activity

```
                           Estimate Std. Error t value Pr(>|t|)    
(Intercept)                -8.99931    1.94474  -4.628 1.71e-05 ***
size                        0.27770    0.16009   1.735   0.0873 .  
females                    -0.34429    0.20501  -1.679   0.0977 .  
diversity                  16.12690    3.72741   4.327 5.08e-05 ***
testosterone                0.07564    0.01633   4.633 1.68e-05 ***
diversity:testosterone     -0.14796    0.03221  -4.594 1.94e-05 ***
```

Activity: thinking about data

Work on the activity (handout).

Activity

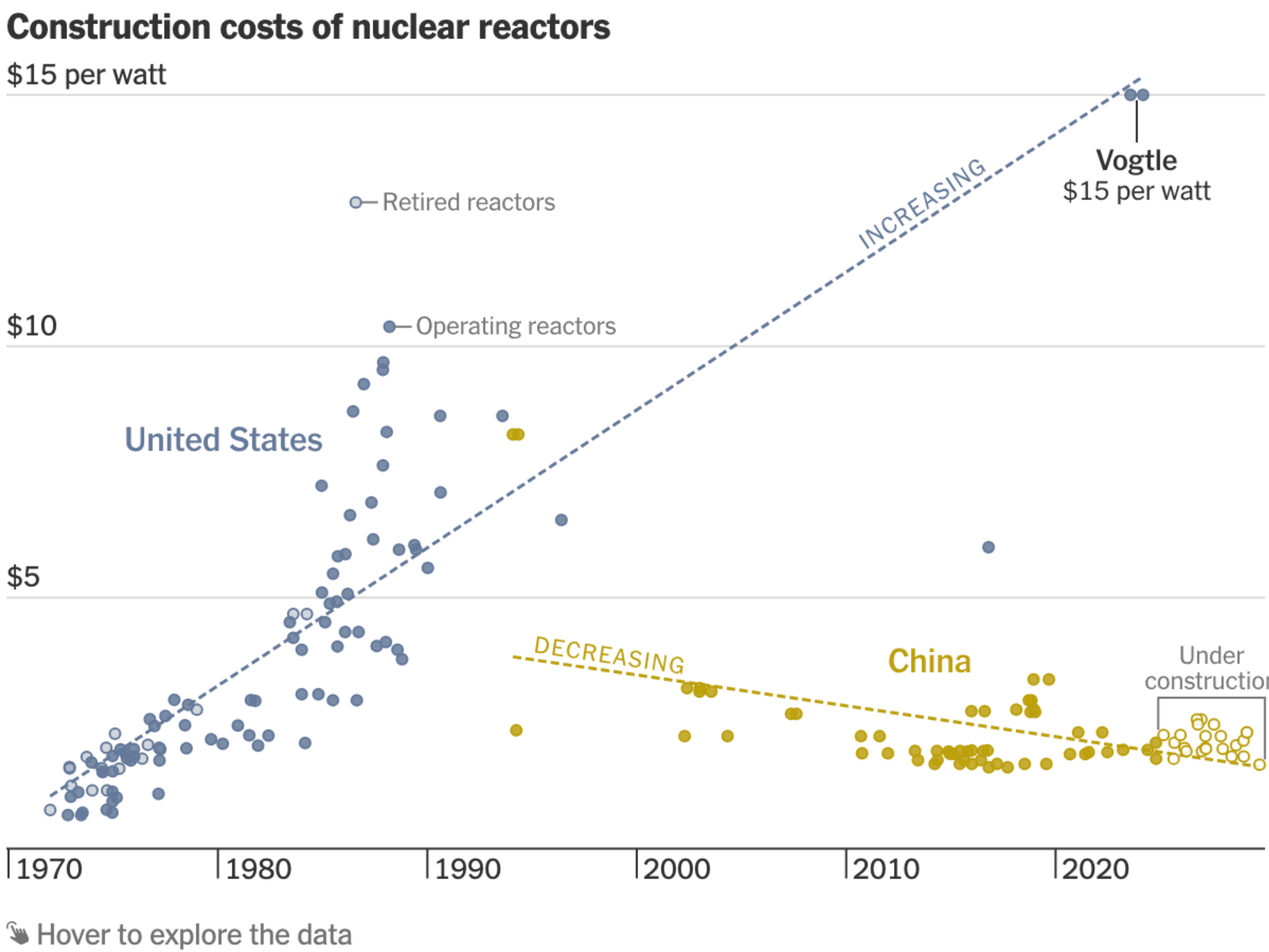
Consider dividing a group into two subgroups. *One* possible measure of group homogeneity is

$$\text{Homogeneity} = \frac{\sum_{j=1}^{p} \sum_{k=1}^{2} m_k (\overline{x}_{\cdot jk} - \overline{x}_{\cdot j \cdot})^2}{\sum_{j=1}^{p} \sum_{k=1}^{2} \sum_{i=1}^{m_k} (x_{ijk} - \overline{x}_{\cdot j \cdot})^2}$$

Activity: thinking about data

Work on the class activity (handout).

Activity



Construction costs of nuclear reactors
$15 per watt
$10
$5
Retired reactors
Operating reactors
United States
INCREASING
Vogtle
$15 per watt
DECREASING
China
Under
construction
1970
1980
1990
2000
2010
2020
Hover to explore the data